gray matter
white matter
Parkinson's disease
Basal Ganglia
Substantia Nigra
Broca's Area
Wernicke's Area
Tan
Frontal lobe
Fluent Aphasia
Expressive Aphasia

Nernst potential
Ionic concentration gradient
Membrane voltage
diffusive/chemical/entropic vs. electrical force/work
Na+/K+ ATPase
leak channels
resting potential
reversal potential
membrane resistance (units)
membrane capacitance (units)
membrane current (units)
membrane time constant (units, typical range)
inhibitory/excitatory post-synaptic potential
membrane integration (role of R, C, τ)
role of ionic charge, concentration differences in reversal potentials

Mistake! Initially flipped

Sodium

sodium Na^+ typical ~~chloride~~ reversal potential 50–65 mV Na^+
typical potassium reversal potential -100--80 mV
chloride Cl^- typical ~~sodium~~ reversal potential -80--60 mV Cl^-
typical resting potential -75--60 mV

Chloride!

Action potential
threshold
positive feedback
upswing, depolarisation
repolarisation
hyperpolarisation
relative refractory period
absolute refractory period
which ions flow, in which directions, in each step

Hodgkin Huxley (memorise mathematical model with physiological interpretations)
voltage-gated ionic conductances
voltage gates
maximal conductances
voltage-gated sodium m gate (role in spiking)
voltage-gated sodium h gate (role in spiking)
voltage-gated potassium n gate (role in spiking)
voltage clamp
patch clamp (cell attached)
whole cell
stable, scalar, linear, first-order ODEs
steady state (t→∞) solutions
time constants
exponential solution ansatz
Forward Euler integration
Forward exponential Euler integration
two compartment models
axial conductance, currents
there are some HH deep thought questions!

LIF (memorise mathematical model with physiological interpretations)
find its threshold
adding a refractory period
derive relationship between spiking frequency/period and applied current
A-level differential calculus, critical thought

soma
dendrite
axon
synapse
axonal terminal / axonal boutaon / presynapse
dendritic spine / postsynapse
action potential propagation (no PDE math)
steps in synaptic transmission
calcium influx in presynapse
readily releasable pool
vesicle fusion
neurotransmitter release and diffusion
neurotransmitter binding
ligand vs voltage-gated ion channels
metabotropic vs ionotropic neurotransmitter receptors
short-term depression/facilitation (memorise mathematical model with physiological interpretations)

Dale's principle
Modulators: Serotonin, histamine, dopamine, adrenaline/norepinephrine, (acetylcholine?)
AMPA receptors ionotropic, glutamate, non-selective cation channel, reversal near 0 mV
formally a mistake: mostly & historically ionotropic. Now maybe partly metabotropic in some senses? Anyway! NMDA: Ionotropic or Metabotropic is NOT on the exam so this ambiguity can be put aside for now.
NMDA receptors ~~metabotropic~~, glutamate, Ca2+ channel, coincidence detector using Mg2+ block), reversal near 0 mV <- Nitrous oxide, Dextrorphan, Ketamine
GABA_A receptor is ionotropic chloride channel (-60 to -80 mV)
GABA_B receptor is metabotropic channel (usually activating a potassium current) (-80 to -100 mV) <- Benzodiazepines
Muscarinic acetylcholine receptor (mAChR) metabotropic (various effects)
Nicotinic acetylcholine receptor (nAChR) non-selective cation channel, reversal near 0 mV <- Nicotine

Hebbian learning
Oja's rule
Finding fixed points and checking stability in nonlinear ODEs in one variable
LTP, LTD, STDP
gradient descent parameter optimisation, loss function
A-level differential calculus for converting loss functions to gradient descent rules

hippocampus
entorhinal cortex (EC)
hippocampal formation
areas DG, CA3, CA1
granule cells
sparse k-winner-take all pattern separation in DG (function?)
auto-associative (attractor, Hopfield-like) recurrent network in CA3 (function?)
CA1 memory readout, place cells
EC input-output, grid cells
Hopfield networks (memorise mathematical model with physiological interpretations)
What is an attractor network?
Hopfield energy function
Calculating Hopfield weights
Calculating Hopfield state evolution

cerebellum
granule cells
Purkinje cells
climbing fibre error
binary perceptron McCulloch–Pitts (linear threshold) "neuron" (memorise mathematical model with physiological interpretations)
mathematical definition: inputs, weights, Heaviside step threshold, outputs
delta learning rule
Evaluating a perceptron/McCulloch–Pitts unit's output for given input
Updating a perceptron/McCulloch–Pitts unit's weights

photoreceptors
bipolar cells
retinal ganglion cells
center-surround tuning
opponent channels
Gabor function (Gaussian * sinusoid)
simple cells (mostly V1 layer 4, some 6)
complex cells (V1 layer 2/3 and higher)
phase invariance
tuning curve
receptive field
what pathway
where pathway
cortical hierarchy
retinotopy
Olhausen and Field's sparse over-complete basis/dictionary model of simple cells
Wilson-Cowan neural fields
Rate coding (temporal and population senses)
Phase coding (hippocampal theta phase precession
Timing code (inter-aural time delays)